Loan Default Risk Classification for Banking

A machine learning approach to smarter credit decisions — predicting loan defaults before they happen.

AKASH S
ANGEL GEORGE
SRAVANA NAMBIAR

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The Problem Banks Face Every Day

Loan defaults are one of the most costly risks in retail banking. Approving high-risk applicants drives up non-performing assets — but being too conservative means lost revenue and missed opportunities.

Too Permissive

High-risk approvals increase NPAs and erode capital reserves

Too Restrictive

Over-rejection limits business growth and alienates creditworthy applicants

Our Goal

Predict default likelihood to enable balanced, data-driven credit decisions

Why This Project Matters

Reduce Financial Risk

Flag high-risk applicants early to protect the institution's lending portfolio from preventable defaults.

Smarter Approvals

Replace intuition-driven decisions with robust, data-driven credit scoring that improves over time.

Regulatory Compliance

SHAP-based model explainability ensures decisions are transparent, auditable, and aligned with fair lending standards.

Class Imbalance Handling

Defaults are rare but financially devastating — our approach specifically addresses the minority class problem.

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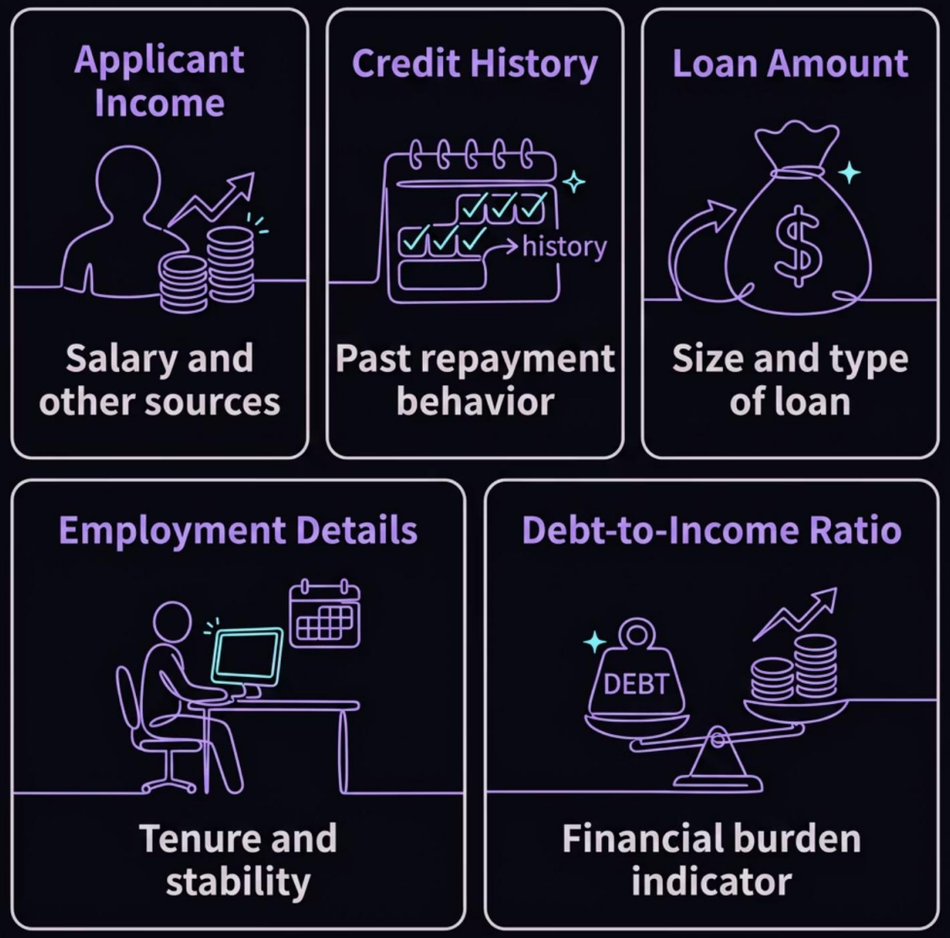
Dataset Overview

Source & Scale

Home Credit Default Risk / LendingClub Dataset — **300,000+ records** capturing real-world loan applicant profiles.

Target Variable

Default (1) vs. **No Default (0)** — highly imbalanced, with defaults representing a critical minority.



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Methodology: From Raw Data to Predictions

Preprocess EDA Features Balance Develop

Each stage was carefully designed to maximize recall — ensuring that high-risk applicants are identified before approval, not after default.

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Models Trained & Compared

Three classifiers were implemented and evaluated head-to-head, with optimization focused on **Recall** to minimize false approvals of risky applicants.

1

Logistic Regression

Class-weighted baseline model. Interpretable and fast — establishes a performance floor for comparison.

2

Random Forest

Ensemble tree-based model. Captures non-linear relationships and reduces overfitting through bagging.

3

LightGBM

Gradient boosting framework optimized for speed and accuracy. Best suited for large, imbalanced tabular datasets.

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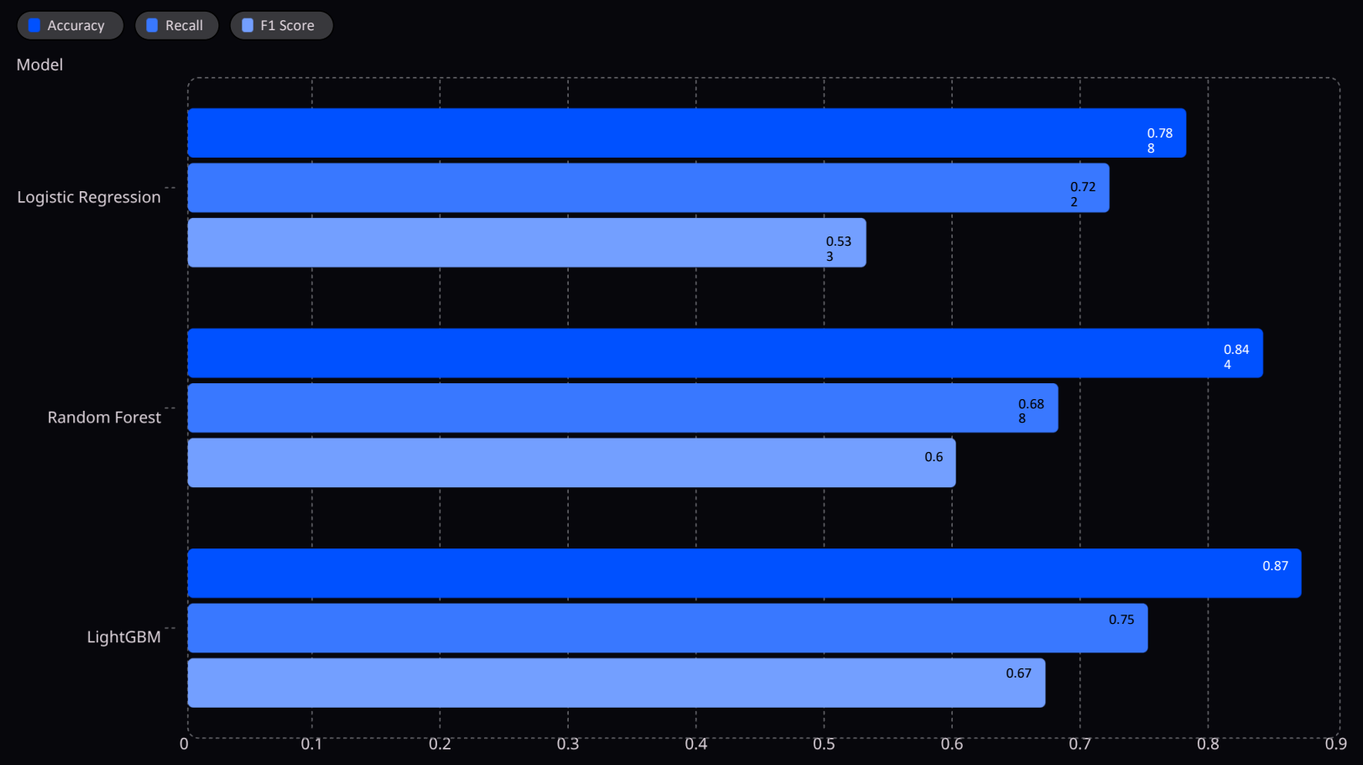
Model Performance Results



Key Takeaways

LightGBM led across all metrics — 87% accuracy, 75% recall, and 0.67 F1 score.

Recall was the primary optimization target: catching true defaults matters more than false precision in lending.

LightGBM: Best Model — AUC 0.7632 | Recall 60.95%

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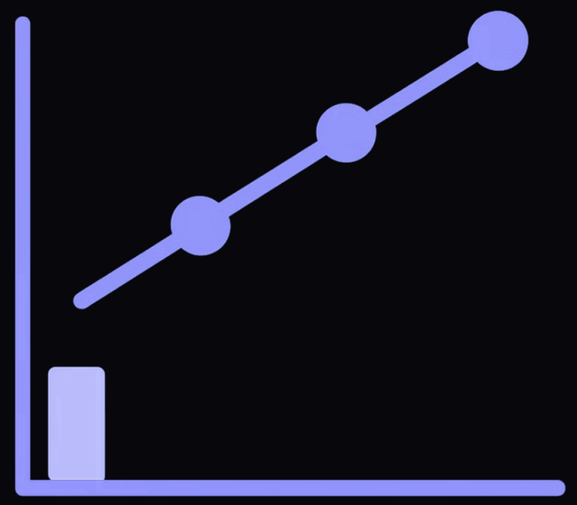
Model Explainability with SHAP



Why Explainability Matters in Banking

Black-box models are insufficient for regulated lending decisions. SHAP (SHapley Additive Explanations) reveals **why** a prediction was made — not just what it is.

Top Default Drivers Identified

- **Income** — Lower income strongly correlates with default risk
- **Credit History** — Past delinquencies are the strongest predictor
- **Loan Amount** — Higher loan-to-income ratios elevate risk scores

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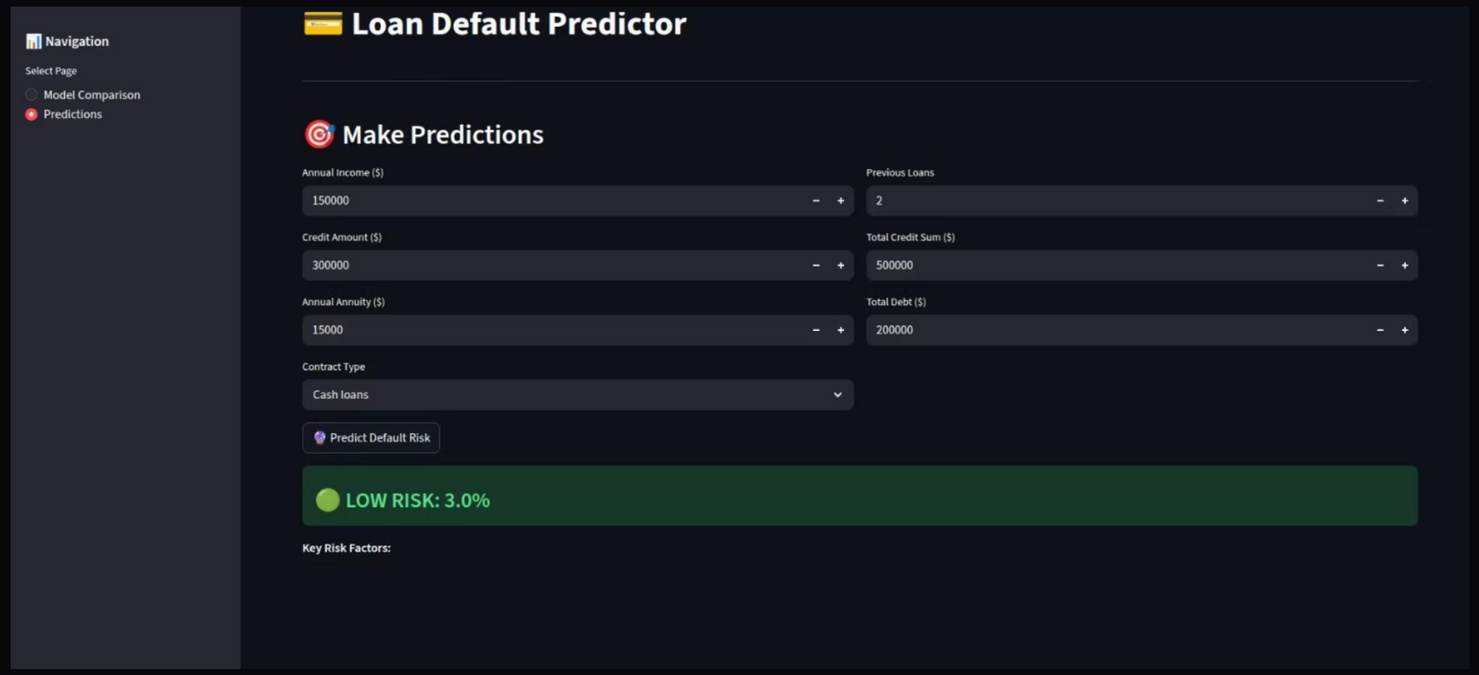
Live Application: Loan Default Predictor

An interactive Streamlit web app allows credit analysts to assess applicant risk in real time — no code required.



Try the live app: akash1231961-loan-default-predictor-app-qbzl8j.streamlit.app

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Summary & Key Outcomes

🏆 Best Model

LightGBM — 87% accuracy, 75% recall, AUC 0.7632

🎯 Recall-First

Optimized to catch risky applicants before costly approvals

🔍 Explainable

SHAP ensures transparent, auditable, and fair lending decisions

🚀 Deployed

Live Streamlit app ready for real-world analyst use